Gnomon Ladders

Stage 0 plan: building the exact angles of the White Queen 600-cell from gnomons (placeholder name)



DICTO · idea 27 October 2026: "square roots could use gnomons" · study 2026-10-27-gnomon-ladders

> **Goal.** Every exact angle found for the 600-cell's six turnings has a cosine √(a + b√5) with a, b rational. Each such number can be built with straightedge and compass. DICTO's idea: build each one from gnomons — the pieces that grow a figure into a larger similar figure (the L-shape that grows a square into the next square; the golden gnomon, the 36°–36°–108° triangle, that grows a golden triangle into the next one). Result wanted: for each angle, a short exact construction ("ladder") made only of gnomon steps, drawn and checked.

What is to be built (inventory, exact)

Turning	Angle	cos²
vertex–edge (whole turn)	18°	5/8 + √5/8
vertex–edge, swaps	9.7323°, 13.6138°	43/88 + 19√5/88; 63/88 + 9√5/88
vertex–triangle (whole turn)	20.9052°	1/2 + √5/6
vertex–triangle, swaps	8.3008°, 13.2825°, 15.4504°, 17.1720°	1/2 + 3√5/14; 1/2 + √5/5; 65/82 + 5√5/82; 22/29 + 2√5/29
vertex–cell (whole turn)	22.2388°	7/16 + 3√5/16
vertex–cell, swaps	7.7612°, 10.3625°, 13.0507°, 15.5225°	9/16 + 3√5/16; 10/19 + 15√5/76; 13/19 + 9√5/76; 23/32 + 3√5/32
edge–triangle (whole turn)	10.8123°	2/3 + 2√5/15
edge–triangle, swaps	5.1524°, 7.6226°	1/2 + 11√5/50; 5/6 + √5/15
edge–cell (whole turn)	13.2825°	1/2 + √5/5
edge–cell, swaps	4.1723°, 7.2560°	22/41 + 42√5/205; 45/58 + 27√5/290
triangle–cell (whole turn)	7.7612°	9/16 + 3√5/16

20 entries, 18 distinct angles (7.7612° and 13.2825° occur twice). Already known by name: 18° (a golden-triangle angle, half the golden gnomon's 36° tip) and 15.5225° (180° minus the 600-cell's dihedral angle; cos = (1 + 3√5)/8).

Stages

Stage 0 — now. This plan; the inventory above (exact, from shadow_turnings_exact.json); prior art; decisions below.

Stage 1 — the moves. Fix a small set of gnomon steps, each exact: (G1) square gnomon ($n^2 \to (n+1)^2$); (G2) golden gnomon (golden triangle → the next, ratio φ); (G3) the right-triangle step that takes a square root (Euclid VI.13, the geometric mean; the Theodorus spiral is a

chain of such gnomon triangles); (G4) adding or halving a length. Prove each step exact in $\mathbb{Q}(\sqrt{5})$ and its square roots.

Stage 2 — the ladders. For each angle, search for the shortest chain of moves that reaches it, starting from a unit and φ; check every chain exactly (each length and angle as a + b√5 or its root). Record the step count. Look for shared rungs: the turnings already share angles, so the ladders may share steps.

Stage 3 — pictures. One drawing per ladder (gnomons in gold on the dark ground, DICTO mark), and one plate with all eighteen. Then a "Gnomon" panel in the White Queen world beside the Shadow view: the current angle drawn as its ladder.

Stage 4 — write-up. Where it goes is DICTO's choice (a section of #63 or its own short paper); asked when Stage 3 is done.

Decisions taken

Placeholder name **Gnomon Ladders**. Exact arithmetic only; floats only for drawing. Start from the two lengths 1 and φ (the 600-cell's own numbers). Heavy searches on dicto-big-brother. Lower-symmetry or approximate constructions are out of scope.

Prior art to check in Stage 0

Gnomons: Euclid, *Elements* II (definition 2) and VI.13; Hero of Alexandria; D'Arcy Thompson, *On Growth and Form* (1917), chapter on the gnomon; the golden gnomon (Penrose tiles; Robinson triangles). Constructibility: Wantzel (1837); constructible numbers $\mathbb{Q}(\sqrt{\ldots})$. The library has no entry under "gnomon" (checked 27 October 2026); web search next.